



# AI Fundamentals & Prompting for TPMs

Week 1 • Day 1 • Customer-Centric Foundations

Technical Product Manager Academy





# Let's Get Started!

Day 1 — AI Fundamentals & Prompting

# Day 1 Objectives

- Describe what a large language model (LLM) can and *cannot* reliably do for a TPM
- Apply four prompt patterns: **Role, Context, Constraints, Format**
- Use AI to accelerate three recurring PM tasks: research, summarization, critique
- Detect and recover from hallucination, grounding failures, and false confidence
- Build a personal **Prompt Pattern Library** with your cohort quad



# Today's Shape

| Clock         | Block                           | Outcome                         |
|---------------|---------------------------------|---------------------------------|
| 09:00 – 09:15 | Opening                         | Meet your quad, ground the week |
| 09:15 – 10:30 | Block 1 + Activity 1            | What LLMs actually are          |
| 10:45 – 12:00 | Block 2 + Activity 2            | Four prompt patterns            |
| 13:00 – 14:15 | Block 3 + Activity 3            | Three TPM uses of AI            |
| 14:30 – 16:00 | Block 4 + Activity 4 + Readouts | Verification & trust            |

# Block 1 — What an LLM Actually Is

Enough mental model to stop being surprised by it

# The Only Mental Model You Need

An LLM is a **very large statistical model of language** that predicts the next token given everything before it. It is not a database. It is not a reasoner. It does not "know" things.

## It is good at

- Rephrasing, summarizing, re-formatting
- Pattern-completing from strong context
- Drafting first passes quickly
- Stress-testing your own writing

## It is bad at

- Giving exact numbers or citations from memory
- Saying "I don't know"
- Reasoning about novel constraints without being shown them
- Telling you when it is wrong



# Three Terms You Will Say Out Loud

## Token

The unit the model reads and writes.

Roughly ~4 characters of English. "Product"  $\approx$  1 token, "Dispatch"  $\approx$  1–2.

## Context window

How many tokens the model can see in one call. Everything outside this window is invisible to it.

## Temperature

A knob from deterministic (0) to creative (1+). PMs usually want low-to-mid for drafts, low for extraction.

**PM implication:** If the model can't see it in the prompt, it's guessing. Always ask yourself: *"What does this model know about my product right now? Nothing."*

# Activity 1 — "Same Question, Three Models"

Quads • 45 minutes

Each quad picks one open PM question (provided list of 10). Each member asks the *same* question to a different tool (ChatGPT, Claude, Gemini/Copilot).

- Compare: Where do the answers agree? Diverge? Hallucinate?
- Score each answer on **usefulness**, **verifiability**, **risk if wrong**
- Prepare a 60-second readout: "We would trust X for Y, and never trust it for Z"



10 min prompt • 15 min compare • 10 min write readout



# Block 2 — Four Prompt Patterns

The smallest scaffold that meaningfully improves output



# The RCCF Scaffold

Every time you ask an AI for non-trivial PM help, include four things: **Role, Context, Constraints, Format**.

| Slot        | Question it answers              | PM example                                                  |
|-------------|----------------------------------|-------------------------------------------------------------|
| Role        | Who is talking?                  | "Act as a senior technical product manager reviewing a PRD" |
| Context     | What does the model need to see? | User persona, the problem, prior attempts                   |
| Constraints | What must be true of the answer? | "Flag any claim that lacks evidence"                        |
| Format      | What shape should it take?       | "Markdown table with columns X, Y, Z"                       |



## Before & After

### ✗ Weak prompt

Write me a user persona for a dispatch app.

*Output is generic, confident, ungrounded, un-actionable.*



### ✓ Scaffolded prompt

Role: Senior TPM for a field-service SaaS.  
Context: Target user = dispatcher at an HVAC company, 50-200 techs, 10+ yrs industry, low tolerance for screens. Pain: re-routing mid-day when a tech calls out sick.  
Constraints: Persona must be grounded in the behaviors above; flag any field that is an assumption I'd need to validate.  
Format: Markdown persona sheet with a final "Assumptions to validate" section.



# Prompting is a Dialogue, Not a Command

- **First pass:** RCCF scaffold
- **Second pass:** "Critique the above as if you were a skeptical engineer"
- **Third pass:** "Now rewrite it, addressing those critiques"
- **Always:** "What in this answer are you least confident about?"

**Caution:** The model will happily invent critiques too. Your judgment is what converts dialogue into value.

## Activity 2 — Rewrite This Prompt

Quads • 50 minutes

Each quad receives three weak prompts taken from the FieldPulse backlog. Rewrite each one using RCCF, then run the original and the rewrite side-by-side.

- Capture: What changed in the output? What did *not* change?
- Add one prompt to your quad's **Prompt Pattern Library**
- Star the prompt your quad would actually reuse on Monday morning

 15 min rewrite • 15 min run • 10 min library entry

# Block 3 — Three Recurring TPM Uses

Where AI actually earns its seat at the table

# Use 1 — Accelerated Research

**The job:** Quickly build context about an unfamiliar domain, user group, or competitor landscape — enough to ask smart questions, not to ship decisions.

## Good fit

- "Explain HVAC dispatcher workflows to me in the language they'd use"
- "What are five things I might be naive about here?"
- "Give me five search queries that will find primary-source material"

## Bad fit

- "What is the market size of dispatch software?" (will hallucinate)
- "Who is the leader?" (outdated and confident)
- "Cite a study that proves..." (fabricated citations)

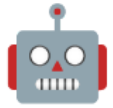


## Use 2 — Summarization with Evidence Tags

When you feed a model a transcript, survey, or support-ticket batch, ask it to **tag each claim with where in the source it came from**. Unsourced claims are the ones to interrogate.

```
Role: Research analyst.  
Context: Below are 14 support tickets from FieldPulse dispatchers.  
Task: Extract the top 5 recurring frustrations.  
Constraints:  
- Each frustration must cite 2+ ticket numbers as evidence  
- Flag any frustration you inferred without direct evidence  
Format: Markdown list; each entry: Frustration / Evidence tickets / Confidence (H/M/L).
```





## Use 3 — Self-Critique Loop

Before sharing a product requirements document (PRD), persona, or problem statement, run a structured critique pass:

1. "Read the following as a skeptical engineer. What three questions will you ask me first?"
2. "Read the following as the stakeholder who most often blocks launches here. What objection worries me most?"
3. "Read the following as the customer in the persona. Where does it not ring true?"

**Why it works:** You get three perspectives in five minutes without socializing a half-baked draft with real people.

## Activity 3 — The Three Hats

Quads • 50 minutes

Each quad takes the FieldPulse one-paragraph problem brief (provided). Using AI, run it through three critique hats: **Engineer**, **Blocker-stakeholder**, **Target customer**.

- Capture the single most uncomfortable question each hat raised
- Decide which one your quad cannot answer yet — that becomes your Day 3 research question



10 min each hat • 10 min synthesize

# Block 4 — Verification & Trust

How grown-up TPMs use AI without being fooled by it



# The Four Failure Modes

| Failure                   | What it looks like                          | Mitigation                                                             |
|---------------------------|---------------------------------------------|------------------------------------------------------------------------|
| <b>Hallucination</b>      | Confident fabricated facts, citations, APIs | Treat outputs as leads, not sources. Verify numerics & names manually. |
| <b>Context starvation</b> | Plausible but generic answer                | Add Role + Context explicitly. Paste in the source material.           |
| <b>Sycophancy</b>         | Agrees with whatever premise you give       | Ask for counter-arguments and rank them by strength.                   |
| <b>Anchoring</b>          | Your framing shapes its output              | Ask for the opposite position as a second turn.                        |



# Evidence Tiers (we'll use this all week)

When you write something down — persona traits, pain points, problem statements — tag each claim with its evidence tier so reviewers can calibrate trust.

| Tier         | Source                                           | Trust                               |
|--------------|--------------------------------------------------|-------------------------------------|
| Observed     | You watched it happen, or recorded data shows it | High                                |
| Reported     | A user said it in an interview or ticket         | Medium                              |
| Inferred     | You deduced it from related evidence             | Low                                 |
| AI-generated | A model produced it — not yet validated          | Lowest; must be upgraded before use |

# Activity 4 — Prompt Pattern Library (Publish)

Quads • 55 minutes • Readouts

Each quad finalizes a shared **Prompt Pattern Library** for use the rest of the academy. Minimum four entries:

- One **Research** prompt
- One **Summarization-with-evidence** prompt
- One **Critique-hat** prompt
- One **"Where could you be wrong?"** prompt

 25 min draft • 10 min test • 10 min quad readout



# Day 1 Wrap

## What we built

- A mental model of what an LLM is (and isn't)
- The RCCF prompt scaffold
- Three reusable TPM uses: research, summarization, critique
- A Prompt Pattern Library per quad

## What opens tomorrow

- **Working Backwards** — we write the press release *before* the product
- Your quad will use today's prompts to draft and critique your first PR/FAQ

**Bring to Day 2:** Your quad's Prompt Pattern Library + one domain you'd like to draft a Working-Backwards PR for.

# Questions & Reflection

Before you leave: what's the one prompt you'll use tomorrow morning?